MIL-T-20177D
19 April 1988
SUPERSEDING
MIL-T-20177C
22 FEBRUARY 1972

MILITARY SPECIFICATION

TRAINING AID, MODEL-BRIDGE, FOR PANEL BRIDGE, M2

BAILEY-TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers model-bridge training aid components for a model of the panel bridge, M2, Bailey-type and the chests for storage of the components.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6910

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:23Z
Check the source to verify that this is the currentversion before use.

SPECIFICATIONS

FEDERAL

T-T-881	-	Twine, Cotton, Seine.
FF-N-836	-	Nut: Square, Hexagon, Cap, Slotted, Castellated, Clinch, Knurled and Welding.
FF-S-92	-	Screws, Machine: Slotted, Cross-Recessed or Hexagon Head.
FF-S-107	-	Screw, Tapping and Drive.
RR-C-271	-	Chains Attachments, Welded, Weldless, and Roller Chain.
TT-S-176	-	Sealer, Surface, Varnish Type, Floor, Wood or Cork.
GGG-P-471	-	Pliers.
GGG-S-121	-	Screwdriver and Screw Starter, Hand.
GGG-W-636	-	Wrenches, Box and Open End (Nonadjustable).
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-W-530	-	Webbing, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-C-12044	-	Chests, Plywood.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

2.1.2 Other Government drawings. The following other Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:23Z
Check the source to verify that this is the current version before use.

DRAWINGS

ME

D7537-1 to -19 (Except 14 and 18)	- Training Aid Bridge Model Kit Panel Bridge M2, Bailey Type.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 21	- Naval Brass Rod, Bar, and Shapes.
B 36	- Brass Plate, Sheet, Strip, and Rolled Bar.
B 85	- Aluminum Alloy Die Castings.
B 134	- Brass Wire.
B 159	- Phosphor Bronze Wire.
B 209	- Aluminum-Alloy Sheet and Plate.
B 211	- Aluminum-Alloy Bars, Rods, and Wire.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The model bridge training aid or training aid components as specified (see 6.2) shall be as shown on Drawings D7537-1 to -13, 15 to 17, 19 and as specified herein. The complete training aid when required shall be furnished disassembled and packaged in chests.

3.1.1 Drawings. The drawings forming a part of this specification are

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:23Z
Check the source to verify that this is the current version before use.

tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of sample training aids or training aids components from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish training aids conforming to this specification.

3.3 Materials. Materials shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Aluminum-alloy bars, rods, and wire. Aluminum-alloy bars, rods, and wire shall conform to ASTM B 211, alloy 5052, temper optional.

3.3.2 Aluminum-alloy die castings. Aluminum-alloy die castings shall conform to ASTM B 85, alloy S12B or S12A.

3.3.3 Aluminum-alloy sheet. Aluminum-alloy sheet shall conform to ASTM B 209, alloy 5052, temper optional.

3.3.4 Brass rods. Brass rods shall conform to ASTM B 21, alloy 485, temper half-hard.

3.3.5 Brass sheets. Brass sheets shall conform to ASTM B 36, alloy 268, temper half-hard.

3.3.6 Brass wire. Brass wire shall conform to ASTM B 134, alloy 270, temper half-hard.

3.3.7 Phosphor-bronze wire. Phosphor-bronze wire shall conform to ASTM B 159, alloy 510, temper soft.

3.3.8 Cotton seine twine. Cotton seine twine shall conform to T-T-881, type II, with a minimum breaking strength of 35 pounds.

3.3.9 Cotton webbing. Cotton webbing shall conform to MIL-W-530, type III, class 1b.

3.3.10 Nuts. Nuts shall conform to FF-N-836 and shall be as specified on the drawings.

3.3.11 Machine screws. Machine screws shall conform to FF-S-92 and be as specified on the drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:23Z
Check the source to verify that this is the current version before use.

3.3.12 Self-tapping screws. Self-tapping screws shall conform to FF-S-107 and be as specified on the drawings.

3.3.13 Chain. Chain shall conform to RR-C-271, type II, class 7, trade number 20.

3.3.14 Wood. Wood for chess may be of any of the following species: Ponderosa pine, sugar pine, or white pine. Wood shall have a moisture content not to exceed 12 percent. Slope of grain shall be not more than 1 in 12. Wood shall be free of defects and knots, except that the following will be permitted:

a. Discoloration due to mineral or pitch streaks or light stain.
b. Surface checks or shakes of barely perceptible openings not more than 4 inches long.

3.3.14.1 Impregnation of wood. Wood for chess shall be impregnated with a phenolic-base compound conforming to TT-S-176, class 1. Impregnation shall be done after fabrication is completed and shall have penetration of not less than 1/16 inch. Excess impregnation compound shall be removed from the surfaces of the wood so as not to form a thick film or lumps on the surface or in corners. Impregnation shall produce minimum discoloration of the wood parts and shall not interfere with subsequent paint coatings which will be applied.

3.4 Assembly tools. Assembly tools shall be as specified herein and as shown on the drawings.

3.4.1 Pliers. Pliers shall conform to GGG-P-471, type XI, class 1, style B, 6-inch normal size.

3.4.2 Screwdrivers. Screwdrivers shall conform to GGG-S-121, type I, class 5, style 2, design A, shape a, 3-inch normal size.

3.4.3 Wrenches. Wrenches shall conform to GGG-W-636, type VI, style 2 with wrench openings as shown on the drawings.

3.5 Chests. The chest shall be as shown on the drawings and as specified in MIL-C-12044.

3.6 Identification marking. The training aid chest shall be marked as shown on the drawings and as specified in MIL-C-12044.

3.7 Treatment and painting. All bridge parts and components of the training aid, except nuts, machine screws, and pins shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, lusterless finish except that the color number designation, shall be 34087 in accordance with FED-STD-595, except as specified in 3.7.1.

3.7.1 Colors. Finish paint color for the items specified in table I shall match color numbers of FED-STD-595.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:23Z
Check the source to verify that this is the current version before use.

TABLE I. Paint Colors.

Item (Dwg. part No.)	Quantity to be painted in each kit	Color	Color No.
7537-2-1	1	Brown	30324
7537-3-1	1	Red	31136
7537-6-1	1	Yellow	33538
7537-6-2	1	Blue	34325
7537-6-3	1	Blue	35231
7537-7-1	1	Gray	35237
7537-7-2	1	Orange	32246

3.8 Workmanship. Glued, cemented, and soldered joints shall make a secure and clean connection.

3.8.1 Casting and forgings. All parts and components of the training aid which include castings and forgings shall be cleaned of harmful extraneous material such as sand, dirt, pits, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect fit or function.

3.8.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of metal shall be done by methods that will not cause injury to the metal. Corners shall be square and true. All bends shall be made with controlled means to insure uniformity of size and shape.

3.8.3 Assembly and interchangeability. Like components and parts shall be fabricated to be duplicates within the tolerance specified, shall be interchangeable, and shall be capable of assembly without the use of any tools or aids other than those included on the drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right tc perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:23Z
Check the source to verify that this is the currentversion before use.

contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article training aid or training aid components, as applicable, shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article test. The first article training aid or training aid components, as applicable, shall be subjected to the tests as specified in 4.5.2. Failure of any test shall be cause for performing the inspection in 4.2b and providing corrective action to eliminate the deficiencies responsible for the failure. Continued failure of any test after corrective action has been taken will be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Lot. A lot for examination shall consist of all like components and parts of the training aids offered for delivery at one time.

4.4.1.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective.

4.4.3 Tests. Each training aid or training aid component, as applicable, shall be tested as specified in 4.5.2.1. The wood used for the chess shall be tested as specified in 4.5.2.2. Failure of either test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The training aid or training aid components shall be examined as specified herein for the following defects:

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:23Z
Check the source to verify that this is the currentversion before use.

101. Material not as specified.
102. Dimensions not as specified.
103. Components and assembly tools missing or not as specified (see 3.4).
104. Castings and forgings not as specified.
105. Workmanship not as specified (see 3.8).
106. Treatment and painting not as specified (see 3.7).
107. Identification marking or stenciling not as specified, missing or illegible (see 3.6).

4.5.2 Test.

4.5.2.1 Assembly and interchangeability. The training aids or training aid components, as applicable, shall be tested for interchangeability of parts, ease of assembly, fit and alignment. Where entire training aids are required, the first article shall be constructed of triple-triple construction, six bays long with two single bay ramps, overhead bracing and footwalks. Where training aid components are required, such assemblies as are applicable shall be made. Any indication of noninterchangeability or difficulty of assembly shall constitute failure of this test.

4.5.2.2 Moisture content. The moisture content of wood for chess shall be determined by use of a resistance type moisture meter. Nonconformance to 3.3.14 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Examination. Each training aid shall be examined for the following defects. Any defects found shall be corrected prior to shipment of the item.

108. Surfaces of components requiring a contact preservative in accordance with the referenced document not coated for level A.
109. Components not placed and secured in the appropriate chest and chest lid closed and secured for level A or C.
110. Components without chests not consolidated as specified for level A (see 5.1.1.2).
111. Skids not applied to shipping container for level A or B.
112. Strapping not zinc coated for level A.
113. Strapping not placed adjacent to the skids for level A or B.
114. Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:23Z
Check the source to verify that this is the current version before use.

5.1.1 Level A. Components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.1 Complete training aid with chest. The component of each complete training aid shall be placed and secured in the appropriate chest and the chest lid closed and secured.

5.1.1.2 Replacement components without chest. The components of like description shall be consolidated together in close-fitting boxes conforming to PPP-B-636, class weather-resistant, grade as applicable, style optional. Cushioning shall be provided to prevent abrasive action between components within the box. Strapping shall not be required.

5.1.2 Level C. Components of each complete training aid shall be placed and secured in the appropriate chest and the chest lid closed and secured. Replacement components without chest shall be preserved to afford protection against deterioration and damage during handling from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Complete training aid with chest. The chests, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-621, class 2, style optional or PPP-B-601, overseas type, style optional. The box shall be modified by the installation of skids and strapping in accordance with the applicable box specification; strapping shall be flat zinc coated and placed adjacent to the skids.

5.2.1.2 Replacement components without chest. Components, preserved as specified in 5.1 shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional. The box shall be modified by the installation of skids and strapping in accordance with the box specification; strapping shall be flat zinc coated and placed adjacent to the skids.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A, except boxes shall be domestic type or class as applicable and strapping need not be zinc coated.

5.2.3 Level C. The chests containing one complete training aid shall be palletized or shall be consolidated into a single, sturdy container having strength appropriate for the size and weight of the chests with contents. Chests shall be blocked to prevent movement within the container. If palletized, the chests shall be secured to the pallet with flat steel strapping. Replacement components without chests shall be packed in sturdy, snug fitting containers of wood, plywood, or fiberboard. Containers shall be closed securely to prevent loss of components.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:23Z
Check the source to verify that this is the current version before use.

5.3 Marking. In addition to any special marking required by the contract or order, chests, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The training aid is intended for use in training troops in the assembly of the M2 Bailey-type panel bridge.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Whether a complete training aid or components are required (see 3.1).
c. When a first article is not required for inspection and approval and the number of units required (see 3.2).
d. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the items should be a training aid or training aid components. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition if the first article.

6.4 Provisioning. The contracting officer should include provisioning requirements for assembly tools and instructions on shipment of training aids. A suggested paragraph is as follows:

"Shipment of training aids shall include assembly tools and instructions unless exceptions are provided elsewhere in the contract."

6.5 Subject term (key word) listing.
Bailey bridge
Chess
Model bridge
Panel bridge
Training aid

6.6 Changes from precious issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD

User activity:
Navy - MC

Preparing activity:
ME

Project 6910-0010

☆U.S. GOVERNMENT PRINTING OFFICE: 1988 - 505-033/91553

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:23Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
U.S. Army Belvoir RD&E Center
ATTN: STRBE-TSE
Fort Belvoir, VA 22060-5606

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:23Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-20177D	Training Aid, Model-Bridge, For Panel Bridge, M2, Bailey Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:23Z
Check the source to verify that this is the current version before use.